2026-01-31 - TRAFFIC ANALYSIS EXERCISE ANSWERS

Link to the exercise:

- https://www.malware-traffic-analysis.net/2026/01/31/index.html

Links to some tutorials I've written that should help with this exercise:

- https://www.malware-traffic-analysis.net/tutorials.html

ENVIRONMENT:

- LAN segment range: `10.1.21[.]0/24` (`10.1.21[.]0` through `10.1.21[.]255`)
- Domain: `win11office[.]com`
- AD environment name: `WIN11OFFICE`
- Domain Controller: `10.1.21[.]2 – WIN-LU4L24X3UB7`
- LAN segment gateway: `10.1.21[.]1`
- LAN segment broadcast address: `10.1.21[.]255`

BACKGROUND:

As an analyst at a Security Operations Center (SOC), you check alerts for the past week and find a signature hit for `ET MALWARE Lumma Stealer Victim Fingerprinting Activity` that triggered on traffic from `153.92.1[.]49` over TCP port 80. The alert triggered on 2026-01-27 at 23:05 UTC.

Using the information, you retrieve a packet capture (pcap) of the traffic from the internal IP address that triggered the alert. Based on the pcap, you write up an incident report, so the incident responders can track down the computer and associated user.

TASK:

For this exercise, answer the following questions for your incident report:

- What is the IP address of the infected Windows client?
- What is the MAC address of the infected Windows client?
- What is the host name of the infected Windows client?
- What is the user account name from the infected Windows client?
- What is the full name of the user from the user account?
- What is the domain from `153.92.1[.]49` that triggered the alert for Lumma Stealer?

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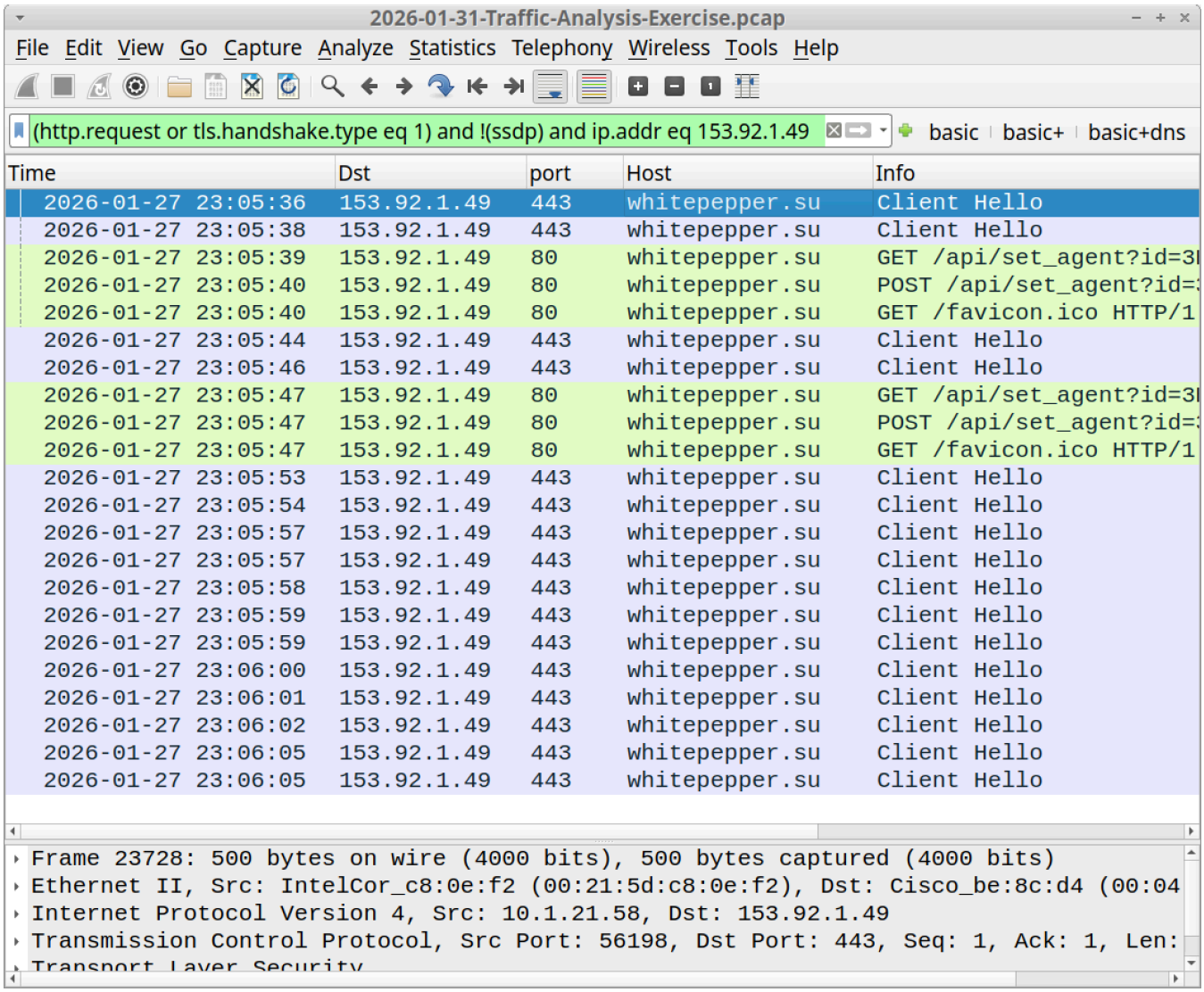
ANSWERS:

Victim Details:

- IP address: `10.1.28[.]58`
- Host name: `DESKTOP-ES9F3ML`
- MAC address: `00:21:5d:c8:0e:f2`
- Windows user account name: `gwyatt`
- Full name of the user: `Gabriel Wyatt`

Lumma Stealer traffic on:

- `153.92.1[.]49:80 – whitepepper[.]su`



Shown above: Filtering in Wireshark to show the traffic on `154.92.1[.]49`.

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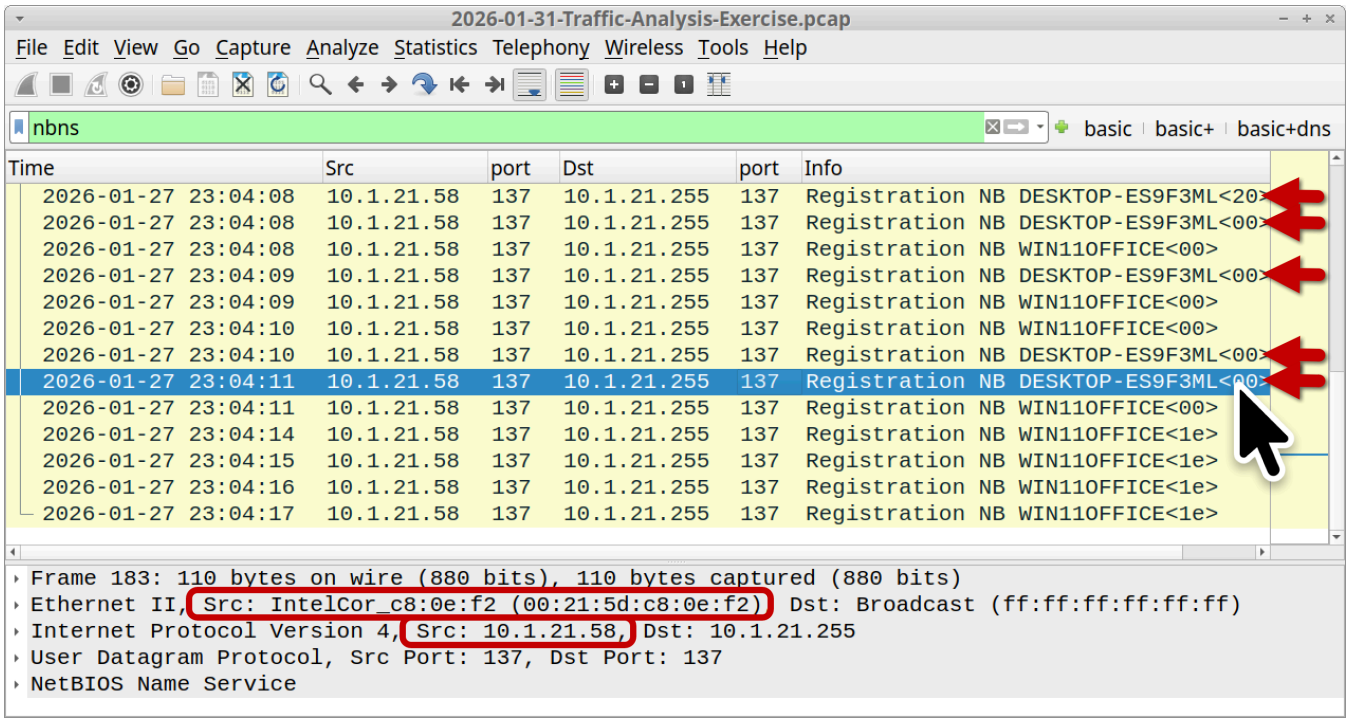
HINTS:

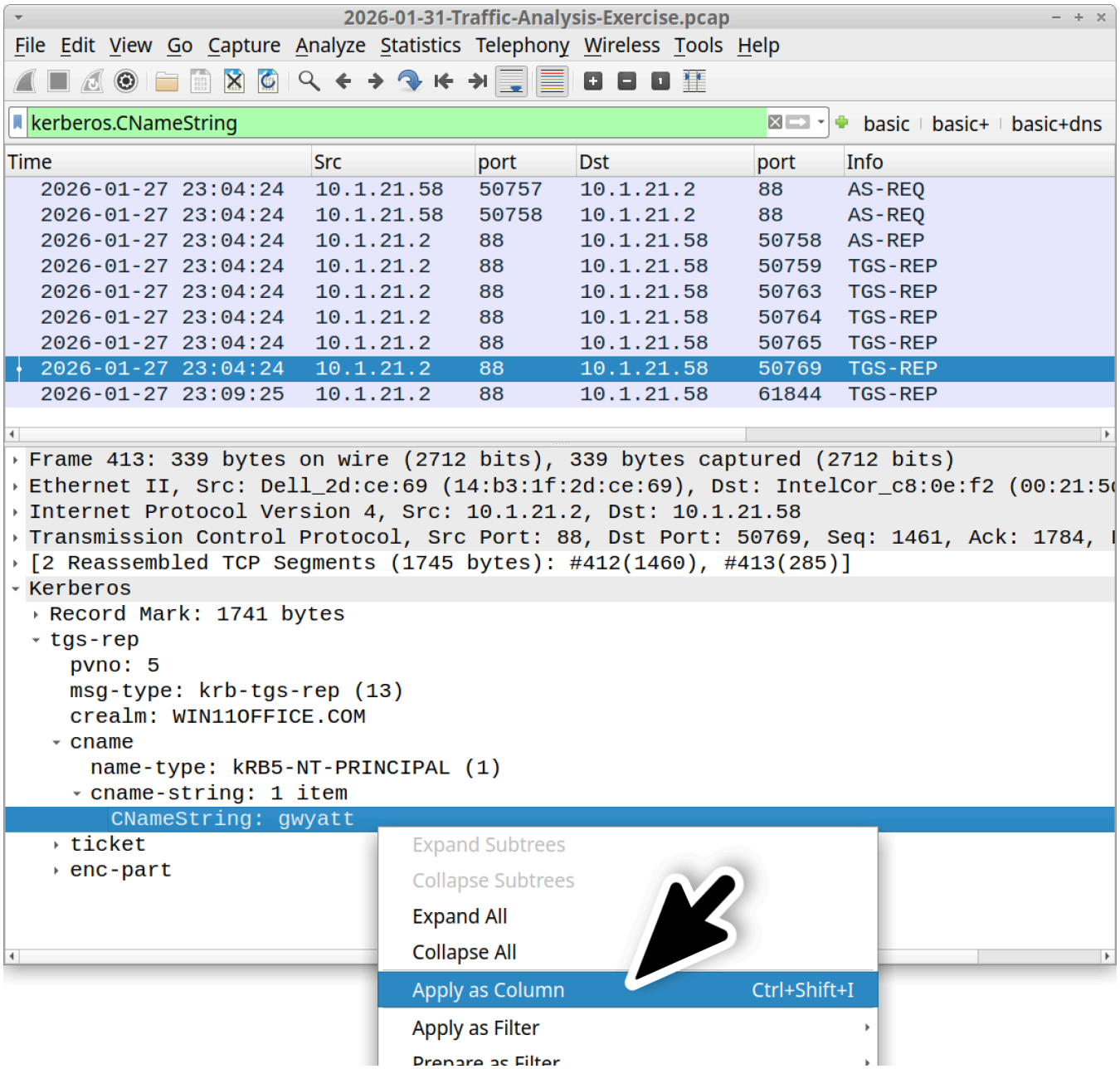
Since this pcap was retrieved from the infected Windows host, we can find that by pivoting on which internal IP address was communicating with `153.92.1[.]49`, which is `10.1.21[.]58`.

Then we can filter on `nbns` in Wireshark to find the host name and view the frame details to find the MAC address.



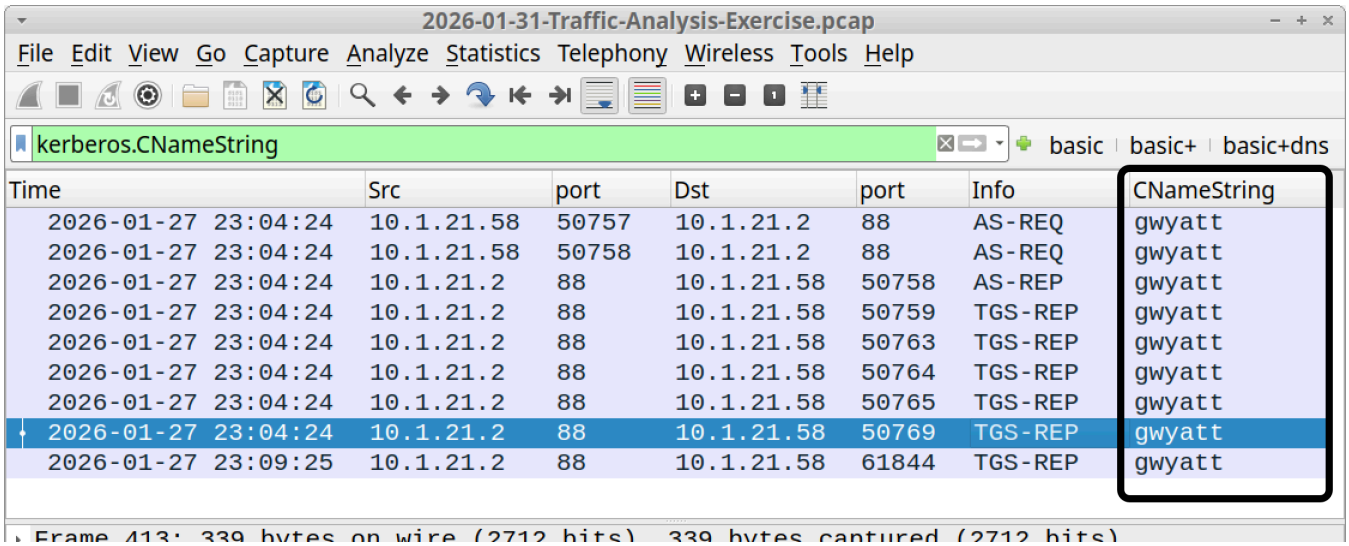
Shown above: Filtering in Wireshark on `nbns` to find the host name.

To find the Windows user account name, filter on `Kerberos.CNameString` and select any of the frames from the results. Then we can drill down into the frame details to find the `CNameString` value, which is `gwyatt`.



Shown above: Filtering to find the `Kerberos.CNameString` value for the user account name.

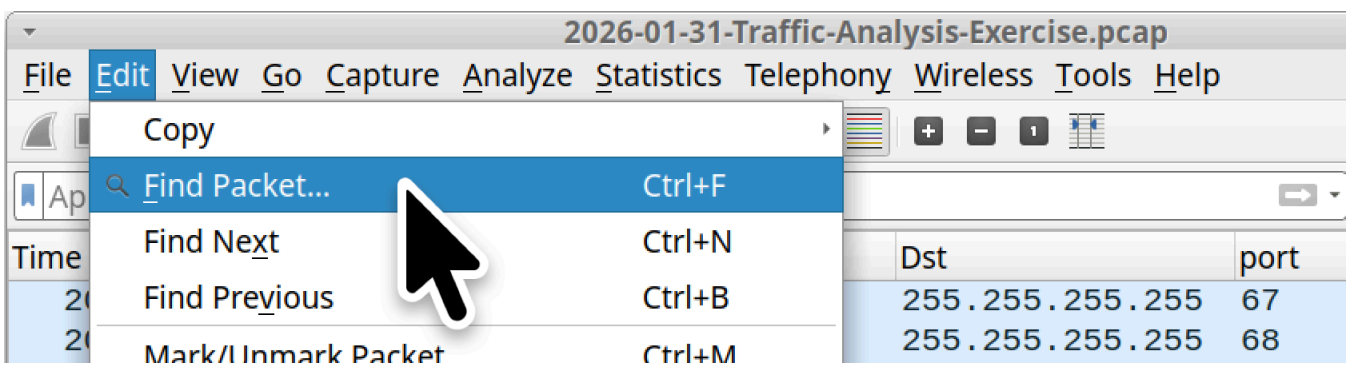
We can apply that `CNameString` field as a column, so we don't have to dig into the frame details each time we want to find the user account name in a pcap.



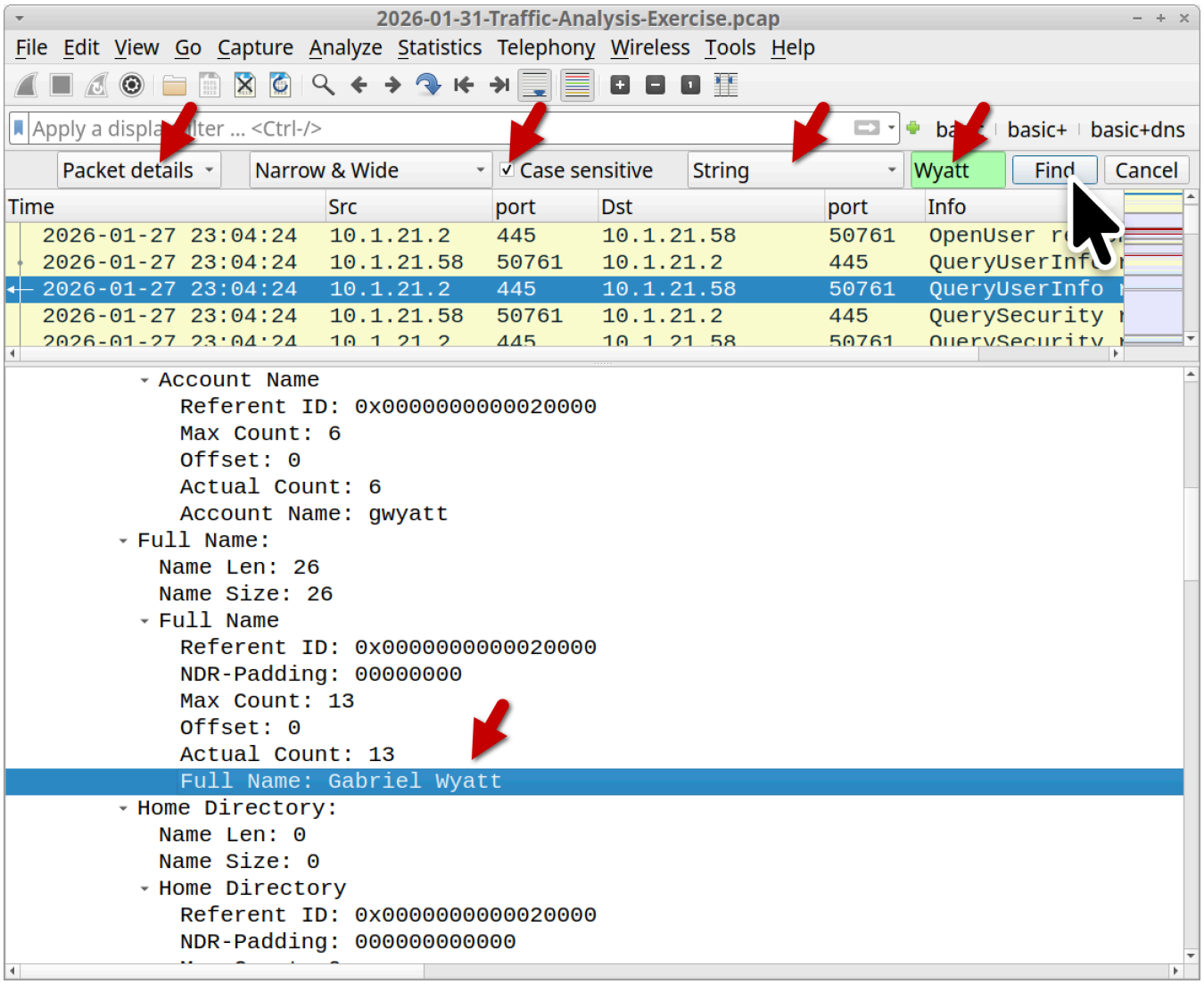
Shown above: The `CnameString` value as a column in our Wireshark display.

Now that we have the user account name, it looks like a first initial last name value, in this case, `g` for the first initial and `wyatt` for the last name.

We can search the packet details for the case-sensitive string `Wyatt` by using the **Find Packet...** function.



*Shown above: The **Find Packet...** function from the **Edit** menu.*



Shown above: Searching packet details to find `Gabriel Wyatt` as the `Full Name` value.

After the Lumma Stealer traffic to `whitepepper[.]su`, the pcap contains HTTPS traffic for two suspicious `.cc` domains:

- `holiday-forever[.]cc`
- `communicationfirewall-security[.]cc`

This follows traffic patterns for follow-up malware I documented on my blog and an ISC diary.

- https://www.malware-traffic-analysis.net/2026/01/01/index.html
- https://www.malware-traffic-analysis.net/2026/01/14/index.html
- https://isc.sans.edu/diary/32628/
- https://www.malware-traffic-analysis.net/2026/01/20/index3.html

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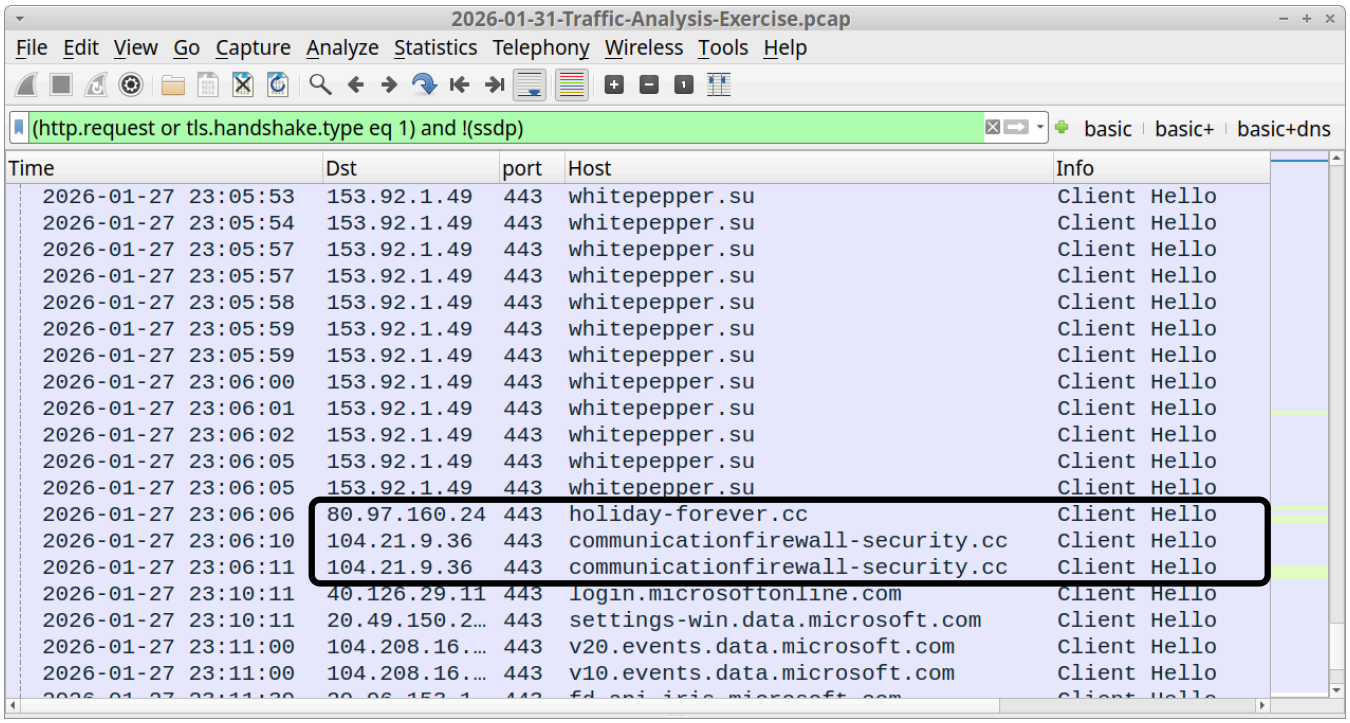

Time	Dst	port	Host	Info
2026-01-27 23:05:53	153.92.1.49	443	whitepepper.su	Client Hello
2026-01-27 23:05:54	153.92.1.49	443	whitepepper.su	Client Hello
2026-01-27 23:05:57	153.92.1.49	443	whitepepper.su	Client Hello
2026-01-27 23:05:57	153.92.1.49	443	whitepepper.su	Client Hello
2026-01-27 23:05:58	153.92.1.49	443	whitepepper.su	Client Hello
2026-01-27 23:05:59	153.92.1.49	443	whitepepper.su	Client Hello
2026-01-27 23:05:59	153.92.1.49	443	whitepepper.su	Client Hello
2026-01-27 23:06:00	153.92.1.49	443	whitepepper.su	Client Hello
2026-01-27 23:06:01	153.92.1.49	443	whitepepper.su	Client Hello
2026-01-27 23:06:02	153.92.1.49	443	whitepepper.su	Client Hello
2026-01-27 23:06:05	153.92.1.49	443	whitepepper.su	Client Hello
2026-01-27 23:06:05	153.92.1.49	443	whitepepper.su	Client Hello
2026-01-27 23:06:06	80.97.160.24	443	holiday-forever.cc	Client Hello
2026-01-27 23:06:10	104.21.9.36	443	communicationfirewall-security.cc	Client Hello
2026-01-27 23:06:11	104.21.9.36	443	communicationfirewall-security.cc	Client Hello
2026-01-27 23:10:11	40.126.29.11	443	login.microsoftonline.com	Client Hello
2026-01-27 23:10:11	20.49.150.2…	443	settings-win.data.microsoft.com	Client Hello
2026-01-27 23:11:00	104.208.16.…	443	v20.events.data.microsoft.com	Client Hello
2026-01-27 23:11:00	104.208.16.…	443	v10.events.data.microsoft.com	Client Hello

Shown above: Two suspcioius `.cc` domains after the Lumma Stealer traffic.

That all I have for this month's traffic analysis exercise. See you all next time!